01 · PROBLEM

Indonesian merchants on DOKU are drowning in payment ops.

Every failed payment, refund, and suspicious charge becomes a manual firefight across dashboards, Slack, and inboxes.

TEAM TOKENBURNER · PROJECT LEDGERSOUL · TRACK BEST PAYMENT USE CASE · AGENTHON 2026

FAILED PAYMENTS

Drop silently. Recovery happens hours late, if at all.

SUSPICIOUS CHARGES

Pile up in inboxes with no consistent triage path.

REFUND REQUESTS

Get approved on gut feel, with no audit trail.

DUPLICATE WEBHOOKS

Re-trigger actions and corrupt reconciliation.

24/7
ops loop

0
audit trail today

hours
recovery delay

3 humans
= 3 different decisions

Payment ops in Indonesia is a 24/7 firefight. Merchants need an autonomous operator, not another dashboard.

LedgerSoul — an autonomous payment-ops agent on DOKU MCP.

A multi-agent system that observes events, reasons about them, calls DOKU tools, verifies results, and writes an immutable audit trail.

LAYER 1 · PAYMENT RAIL & FRAUD

Owned by banks · networks

3DS · Visa Risk · Mastercard DI · AML

LAYER 2 · PAYMENT PLATFORM

Owned by DOKU

Checkout · QRIS · VA · MCP Server

LAYER 3 · PAYMENT OPS (US)

Owned by LedgerSoul

Recovery · triage · refunds · audit

WHY NOT THE DOKU DASHBOARD OR ZAPIER

DOKU DASHBOARD

Manual clicks. Human judgement. No 24/7 loop.

ZAPIER / N8N

If/then glue. No verification. No audit trace.

LEDGERSOUL

Reasoning + tools + verification + audit, every run.

EVERY RUN ENDS IN ONE OF SIX STATES

completed · escalated · duplicate · blocked · failed_verification · error always with a JSON trace

Four agents. One deterministic loop. Real DOKU MCP tools.

Multi-agent reasoning meets a deterministic verification spine — autonomy with proof.

LLM	RULES	TOOLS	CHECKS
REASONING AGENT	**PLANNER AGENT**	**EXECUTOR AGENT**	**VERIFIER AGENT**
classify · extract intent	deterministic plan + policy	TOOL_REGISTRY · DOKU MCP	tool result + audit log

REGISTERED TOOLS · EVERY CALL GOES THROUGH TOOL_REGISTRY

- get_transaction_by_invoice_number
- create_recovery_link
- draft_customer_message
- create_approval_request
- list_doku_mcp_tools
- call_doku_mcp_tool
- write_audit_log

12-STAGE AUTONOMOUS LOOP

boot · observe · validate · idempotency · interpret · plan · policy · act · verify · remember · reflect · stop/escalate

04 · FEATURES & STACK

Built for autonomy, audit, and judge-safe demoability.

Production-shape engineering inside a 12-hour build: tests, lint, judge mode, and real MCP.

KEY FEATURES

- **Multi-agent runtime**
  Reasoning + Planner + Executor + Verifier
- **Real DOKU MCP integration**
  Sandbox MCP server · 35 tools available
- **Explicit tool registry**
  All tool calls auditable, never arbitrary
- **Deterministic verification**
  Every result inspected · no fabrication
- **Idempotency & policy**
  Duplicate webhooks · amount thresholds
- **Judge Mode**
  Locked browser demo · redacted traces
- **Audit trail**
  JSON trace + append-only audit_log.jsonl
- **9 evals · 61 tests**
  All passing · ruff clean

TECH STACK

Language	**Python 3.12**
Server	**FastAPI + Uvicorn**
LLM reasoner	**Pluggable (OpenAI / Claude) via 9router**
Payment rails	**DOKU MCP Server (sandbox)**
Transport	**HTTP JSON-RPC · MCP 2025-06-18**
Persistence	**JSON / JSONL traces · audit log**
Tests / lint	**pytest (61 passing) · ruff**
Deploy	**Docker · Uvicorn · Judge Mode UI**

JUDGE MODE · /judge browser demo · token-protected workflows · redacted traces · /docs · /agent/run · /traces blocked

From hackathon MVP to the operator every DOKU merchant runs.

Direct impact for Indonesian merchants. Clear roadmap from sandbox to merchant-grade.

NOW · v0.1 HACKATHON MVP

- 12-stage autonomous loop
- DOKU MCP sandbox integration
- Judge Mode w/ redacted traces
- 9 evals · 61 tests passing

NEXT · v0.2 PILOT WITH 1 MERCHANT

- Live mode behind explicit flags
- Webhook signature validation
- Slack / Telegram approvals
- SQLite state · richer dashboards

LATER · v1.0 MERCHANT-GRADE

- Multi-merchant tenancy
- Refund + dispute workflows
- Policy editor for ops teams
- LLM-assisted planner upgrades

hours → minutes
recovery cycle

100%
audited runs

0
fabricated decisions

any merchant
with DOKU MCP

WHY DOKU SHOULD CARE

Every DOKU merchant inherits a 24/7 audited operator on day one — without changing checkout, settlement, or fraud rails.